control__eq

control | native_tex_180dpi | no GT-based routing | all models, same input

ACTUAL INPUT

$$3 = 3$$

3=3

REFERENCE RENDERING

$$3 = 3$$

PP-FormulaNet-S

Strict EM: True | Render: rendered | Matrix cells: None

$$3 = 3$$

3=3

PP-FormulaNet_plus-S

Strict EM: True | Render: rendered | Matrix cells: None

$$3 = 3$$

3=3

PP-FormulaNet-L

Strict EM: True | Render: rendered | Matrix cells: None

$$3 = 3$$

3=3

UniMERNet

Strict EM: True | Render: rendered | Matrix cells: None

$$3 = 3$$

3=3

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

control__factorial

control | native_tex_180dpi | no GT-based routing | all models, same input

ACTUAL INPUT

$$3! = 6$$

```
3!=6
```

REFERENCE RENDERING

$$3! = 6$$

PP-FormulaNet-S

Strict EM: True | Render: rendered | Matrix cells: None

```
3!=6
```

$$3! = 6$$

PP-FormulaNet_plus-S

Strict EM: True | Render: rendered | Matrix cells: None

```
3!=6
```

$$3! = 6$$

PP-FormulaNet-L

Strict EM: True | Render: rendered | Matrix cells: None

```
3!=6
```

$$3! = 6$$

UniMERNet

Strict EM: True | Render: rendered | Matrix cells: None

```
3!=6
```

$$3! = 6$$

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

control__indexed_factorial

control | native_tex_180dpi | no GT-based routing | all models, same input

ACTUAL INPUT

$$|S_3| = 3! = 6$$

```
|S_3|=3!=6
```

REFERENCE RENDERING

$$|S_3| = 3! = 6$$

PP-FormulaNet-S

Strict EM: False | Render: rendered | Matrix cells: None

```
|S_{3}|=3!=6
```

$$|S_3| = 3! = 6$$

PP-FormulaNet_plus-S

Strict EM: False | Render: rendered | Matrix cells: None

```
|S_{3}|=3!=6
```

$$|S_3| = 3! = 6$$

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: None

```
|S_{3}|=3!=6
```

$$|S_3| = 3! = 6$$

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: None

```
|S_{3}|=3!=6
```

$$|S_3| = 3! = 6$$

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

control__general_factorial

control | native_tex_180dpi | no GT-based routing | all models, same input

ACTUAL INPUT

$$|S_n| = n!$$

```
|S_n|=n!
```

REFERENCE RENDERING

$$|S_n| = n!$$

PP-FormulaNet-S

Strict EM: False | Render: rendered | Matrix cells: None

```
|S_{n}|=n!
```

$$|S_n| = n!$$

PP-FormulaNet_plus-S

Strict EM: False | Render: rendered | Matrix cells: None

```
|S_{n}|=n!
```

$$|S_n| = n!$$

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: None

```
|S_{n}|=n!
```

$$|S_n| = n!$$

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: None

```
|S_{n}|=n!
```

$$|S_n| = n!$$

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

control__frac

control | native_tex_180dpi | no GT-based routing | all models, same input

ACTUAL INPUT



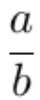
REFERENCE RENDERING



```
\frac{a}{b}
```

PP-FormulaNet-S

Strict EM: True | Render: rendered | Matrix cells: None



```
\frac{a}{b}
```

PP-FormulaNet_plus-S

Strict EM: True | Render: rendered | Matrix cells: None



```
\frac{a}{b}
```

PP-FormulaNet-L

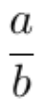
Strict EM: True | Render: rendered | Matrix cells: None



```
\frac{a}{b}
```

UniMERNet

Strict EM: True | Render: rendered | Matrix cells: None



```
\frac{a}{b}
```

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

control__frac_sum

control | native_tex_180dpi | no GT-based routing | all models, same input

ACTUAL INPUT

$$\frac{a+b}{c-d}$$

```
\frac{a+b}{c-d}
```

REFERENCE RENDERING

$$\frac{a+b}{c-d}$$

PP-FormulaNet-S

Strict EM: True | Render: rendered | Matrix cells: None

```
\frac{a+b}{c-d}
```

$$\frac{a+b}{c-d}$$

PP-FormulaNet_plus-S

Strict EM: True | Render: rendered | Matrix cells: None

```
\frac{a+b}{c-d}
```

$$\frac{a+b}{c-d}$$

PP-FormulaNet-L

Strict EM: True | Render: rendered | Matrix cells: None

```
\frac{a+b}{c-d}
```

$$\frac{a+b}{c-d}$$

UniMERNet

Strict EM: True | Render: rendered | Matrix cells: None

```
\frac{a+b}{c-d}
```

$$\frac{a+b}{c-d}$$

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

control__nested_frac

control | native_tex_180dpi | no GT-based routing | all models, same input

ACTUAL INPUT

$$\frac{1}{1+\frac{1}{x}}$$

```
\frac{1}{1+\frac{1}{x}}
```

REFERENCE RENDERING

$$\frac{1}{1+\frac{1}{x}}$$

PP-FormulaNet-S

Strict EM: True | Render: rendered | Matrix cells: None

$$\frac{1}{1+\frac{1}{x}}$$

```
\frac{1}{1+\frac{1}{x}}
```

PP-FormulaNet_plus-S

Strict EM: True | Render: rendered | Matrix cells: None

$$\frac{1}{1+\frac{1}{x}}$$

```
\frac{1}{1+\frac{1}{x}}
```

PP-FormulaNet-L

Strict EM: True | Render: rendered | Matrix cells: None

$$\frac{1}{1+\frac{1}{x}}$$

```
\frac{1}{1+\frac{1}{x}}
```

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: None

$$\frac{1}{1+\frac{1}{x}}$$

```
\frac{1}{1+{\frac{1}{x}}}
```

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

control__long_fraction

control | native_tex_180dpi | no GT-based routing | all models, same input

ACTUAL INPUT

$$y=\frac{3-x}{2}=\frac{3-\frac{1}{3}}{2}$$

```
y=\frac{3-x}{2}=\frac{3-\frac{1}{3}}{2}
```

REFERENCE RENDERING

$$y=\frac{3-x}{2}=\frac{3-\frac{1}{3}}{2}$$

PP-FormulaNet-S

Strict EM: False | Render: rendered | Matrix cells: None

```
y={\frac{3-x}{2}}={\frac{3-{\frac{1}{3}}}{2}}
```

$$y={\frac{3-x}{2}}={\frac{3-{\frac{1}{3}}}{2}}$$

PP-FormulaNet_plus-S

Strict EM: True | Render: rendered | Matrix cells: None

```
y=\frac{3-x}{2}=\frac{3-\frac{1}{3}}{2}
```

$$y=\frac{3-x}{2}=\frac{3-\frac{1}{3}}{2}$$

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: None

```
y={\frac{3-x}{2}}={\frac{3-{\frac{1}{3}}}{2}}
```

$$y={\frac{3-x}{2}}={\frac{3-{\frac{1}{3}}}{2}}$$

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: None

```
y={\frac{3-x}{2}}={\frac{3-{\frac{1}{3}}}{2}}
```

$$y={\frac{3-x}{2}}={\frac{3-{\frac{1}{3}}}{2}}$$

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

control__vec2

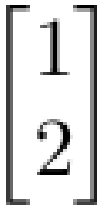
control | native_tex_180dpi | no GT-based routing | all models, same input

ACTUAL INPUT

$$\begin{bmatrix}1\\2\end{bmatrix}$$

```
\begin{bmatrix}1\\2\end{bmatrix}
```

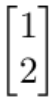
REFERENCE RENDERING



PP-FormulaNet-S

Strict EM: False | Render: rendered | Matrix cells: True

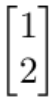
```
\left[\begin{matrix}{1}\\ {2}\\ \end{matrix}\right]
```



PP-FormulaNet_plus-S

Strict EM: False | Render: rendered | Matrix cells: True

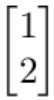
```
\left[\begin{matrix}{1}\\ {2}\\ \end{matrix}\right]
```



PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: True

```
\left[\begin{matrix}{1}\\ {2}\\ \end{matrix}\right]
```



UniMERNet

Strict EM: False | Render: rendered | Matrix cells: False

```
\binom{1}{2}
```



Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

control__vec3

control | native_tex_180dpi | no GT-based routing | all models, same input

ACTUAL INPUT

$$\begin{bmatrix}1\\2\\3\end{bmatrix}$$

```
\begin{bmatrix}1\\2\\3\end{bmatrix}
```

REFERENCE RENDERING

$$\begin{bmatrix}1\\2\\3\end{bmatrix}$$

PP-FormulaNet-S

Strict EM: False | Render: rendered | Matrix cells: True

```
\left[\begin{matrix}{1}\\ {2}\\ {3}\\ \end{matrix}\right]
```

$$\left[\begin{matrix}{1}\\ {2}\\ {3}\\ \end{matrix}\right]$$

PP-FormulaNet_plus-S

Strict EM: False | Render: rendered | Matrix cells: True

```
\left[\begin{matrix}{1}\\ {2}\\ {3}\\ \end{matrix}\right]
```

$$\left[\begin{matrix}{1}\\ {2}\\ {3}\\ \end{matrix}\right]$$

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: True

```
\left[\begin{matrix}{1}\\ {2}\\ {3}\\ \end{matrix}\right]
```

$$\left[\begin{matrix}{1}\\ {2}\\ {3}\\ \end{matrix}\right]$$

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: False

```
\binom{1}{2}
```

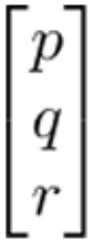
$$\binom{1}{2}$$

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

control__vec_symbolic

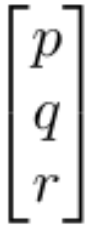
control | native_tex_180dpi | no GT-based routing | all models, same input

ACTUAL INPUT



```
\begin{bmatrix}p\\q\\r\end{bmatrix}
```

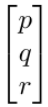
REFERENCE RENDERING



PP-FormulaNet-S

Strict EM: False | Render: rendered | Matrix cells: True

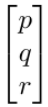
```
\left[\begin{matrix}{p}\\ {q}\\ {r}\\ \end{matrix}\right]
```



PP-FormulaNet_plus-S

Strict EM: False | Render: rendered | Matrix cells: True

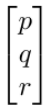
```
\left[\begin{matrix}{p}\\ {q}\\ {r}\\ \end{matrix}\right]
```



PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: True

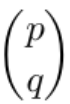
```
\left[\begin{matrix}{p}\\ {q}\\ {r}\\ \end{matrix}\right]
```



UniMERNet

Strict EM: False | Render: rendered | Matrix cells: False

```
\binom{p}{q}
```



Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

control__matrix22

control | native_tex_180dpi | no GT-based routing | all models, same input

ACTUAL INPUT

$$\begin{bmatrix}1 & 2\\3 & 4\end{bmatrix}$$

```
\begin{bmatrix}1&2\\3&4\end{bmatrix}
```

REFERENCE RENDERING

$$\begin{bmatrix}1 & 2\\3 & 4\end{bmatrix}$$

PP-FormulaNet-S

Strict EM: False | Render: rendered | Matrix cells: True

```
\left[\begin{matrix}{1}&{2}\\ {3}&{4}\\ \end{matrix}\right]
```

$$\left[\begin{matrix}{1}&{2}\\ {3}&{4}\\ \end{matrix}\right]$$

PP-FormulaNet_plus-S

Strict EM: False | Render: rendered | Matrix cells: True

```
\left[\begin{matrix}{1}&{2}\\ {3}&{4}\\ \end{matrix}\right]
```

$$\left[\begin{matrix}{1}&{2}\\ {3}&{4}\\ \end{matrix}\right]$$

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: True

```
\left[\begin{matrix}{1}&{2}\\ {3}&{4}\\ \end{matrix}\right]
```

$$\left[\begin{matrix}{1}&{2}\\ {3}&{4}\\ \end{matrix}\right]$$

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: False

```
\begin{array}{r}{\left[1\quad2\right]}\\ {3\quad4\big]}\end{array}
```

$$\begin{array}{r}{\left[1\quad2\right]}\\ {3\quad4\big]}\end{array}$$

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

control__matrix23

control | native_tex_180dpi | no GT-based routing | all models, same input

ACTUAL INPUT

$$\begin{bmatrix}1&2&3\\4&5&6\end{bmatrix}$$

```
\begin{bmatrix}1&2&3\\4&5&6\end{bmatrix}
```

REFERENCE RENDERING

$$\begin{bmatrix}1&2&3\\4&5&6\end{bmatrix}$$

PP-FormulaNet-S

Strict EM: False | Render: rendered | Matrix cells: True

```
\left[\begin{matrix}{1}&{2}&{3}\\ {4}&{5}&{6}\\ \end{matrix}\right]
```

$$\left[\begin{matrix}{1}&{2}&{3}\\ {4}&{5}&{6}\\ \end{matrix}\right]$$

PP-FormulaNet_plus-S

Strict EM: False | Render: rendered | Matrix cells: True

```
\left[\begin{matrix}{1}&{2}&{3}\\ {4}&{5}&{6}\\ \end{matrix}\right]
```

$$\left[\begin{matrix}{1}&{2}&{3}\\ {4}&{5}&{6}\\ \end{matrix}\right]$$

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: True

```
\left[\begin{matrix}{1}&{2}&{3}\\ {4}&{5}&{6}\\ \end{matrix}\right]
```

$$\left[\begin{matrix}{1}&{2}&{3}\\ {4}&{5}&{6}\\ \end{matrix}\right]$$

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: False

```
\begin{array}{r}{\left[1\quad2\quad3\right]}\\ {4\quad5\quad6\big]}\end{array}
```

$$\begin{array}{r}{\left[1\quad2\quad3\right]}\\ {4\quad5\quad6\big]}\end{array}$$

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

control__matrix33

control | native_tex_180dpi | no GT-based routing | all models, same input

ACTUAL INPUT

$$\begin{bmatrix}1&2&3\\4&5&6\\7&8&9\end{bmatrix}$$

```
\begin{bmatrix}1&2&3\\4&5&6\\7&8&9\end{bmatrix}
```

REFERENCE RENDERING

$$\begin{bmatrix}1&2&3\\4&5&6\\7&8&9\end{bmatrix}$$

PP-FormulaNet-S

Strict EM: False | Render: rendered | Matrix cells: True

```
\left[\begin{matrix}{1}&{2}&{3}\\ {4}&{5}&{6}\\ {7}&{8}&{9}\\ \end{matrix}\right]
```

$$\left[\begin{matrix}{1}&{2}&{3}\\ {4}&{5}&{6}\\ {7}&{8}&{9}\\ \end{matrix}\right]$$

PP-FormulaNet_plus-S

Strict EM: False | Render: rendered | Matrix cells: True

```
\left[\begin{matrix}{1}&{2}&{3}\\ {4}&{5}&{6}\\ {7}&{8}&{9}\\ \end{matrix}\right]
```

$$\left[\begin{matrix}{1}&{2}&{3}\\ {4}&{5}&{6}\\ {7}&{8}&{9}\\ \end{matrix}\right]$$

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: True

```
\left[\begin{matrix}{1}&{2}&{3}\\ {4}&{5}&{6}\\ {7}&{8}&{9}\\ \end{matrix}\right]
```

$$\left[\begin{matrix}{1}&{2}&{3}\\ {4}&{5}&{6}\\ {7}&{8}&{9}\\ \end{matrix}\right]$$

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: True

```
\left[\begin{array}{l l l}{1}&{2}&{3}\\ {4}&{5}&{6}\\ {7}&{8}&{9}\end{array}\right]
```

$$\left[\begin{array}{l l l}{1}&{2}&{3}\\ {4}&{5}&{6}\\ {7}&{8}&{9}\end{array}\right]$$

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

control__augmented

control | native_tex_180dpi | no GT-based routing | all models, same input

ACTUAL INPUT

$$\left[\begin{array}{cc|c}1&-3&0\\2&4&0\end{array}\right]$$

```
\left[\begin{array}{cc|c}1&-3&0\\2&4&0\end{array}\right]
```

REFERENCE RENDERING

$$\left[\begin{array}{cc|c}1&-3&0\\2&4&0\end{array}\right]$$

PP-FormulaNet-S

Strict EM: False | Render: unsupported | Matrix cells: False

```
\left[\begin{array}{c c|c}{1}&{-3}&{0}\\ {2}&{4}&{0}\ }\end{array}\right]
```

PP-FormulaNet_plus-S

Strict EM: False | Render: rendered | Matrix cells: True

$$\left[\begin{array}{c c||c}{1}&{-3}&{0}\\ {2}&{4}&{0}\\ \end{array}\right]$$

```
\left[\begin{array}{c c||c}{1}&{-3}&{0}\\ {2}&{4}&{0}\\ \end{array}\right]
```

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: True

$$\left[\begin{array}{c c|c}{1}&{-3}&{0}\\ {2}&{4}&{0}\end{array}\right]$$

```
\left[{\begin{array}{c c|c}{1}&{-3}&{0}\\ {2}&{4}&{0}\end{array}}\right]
```

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: False

$$\left[\begin{array}{l l|l}{1}&{-3}\\ {2}&{~4}\end{array}\middle|\begin{array}{l}{0}\\ {0}\end{array}\right]$$

```
\left[\begin{array}{l l|l}{1}&{-3}\\ {2}&{~4}\end{array}\middle|\begin{array}{l}{0}\\ {0}\end{array}\right]
```

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

control__line1

control | native_tex_180dpi | no GT-based routing | all models, same input

ACTUAL INPUT

$$3x = 1$$

```
3x=1
```

REFERENCE RENDERING

$$3x = 1$$

PP-FormulaNet-S

Strict EM: True | Render: rendered | Matrix cells: None

$$3x = 1$$

```
3x=1
```

PP-FormulaNet_plus-S

Strict EM: True | Render: rendered | Matrix cells: None

$$3x = 1$$

```
3x=1
```

PP-FormulaNet-L

Strict EM: True | Render: rendered | Matrix cells: None

$$3x = 1$$

```
3x=1
```

UniMERNet

Strict EM: True | Render: rendered | Matrix cells: None

$$3x = 1$$

```
3x=1
```

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

control__lines2

control | native_tex_180dpi | no GT-based routing | all models, same input

ACTUAL INPUT

$$\begin{aligned}3x &= 1\\ x+2y &= 3\end{aligned}$$

```
\begin{aligned}3x&=1\\x+2y&=3\end{aligned}
```

REFERENCE RENDERING

$$\begin{aligned}3x &= 1\\ x+2y &= 3\end{aligned}$$

PP-FormulaNet-S

Strict EM: False | Render: rendered | Matrix cells: None

```
\begin{array}{r}{3x=1}\\ {x+2y=3}\end{array}
```

$$\begin{array}{r}{3x=1}\\ {x+2y=3}\end{array}$$

PP-FormulaNet_plus-S

Strict EM: False | Render: rendered | Matrix cells: None

```
\begin{array}{r}{{x}=1,}\\ {{x}+2{y}=3.}\\ \end{array}
```

$$\begin{array}{r}{{x}=1,}\\ {{x}+2{y}=3.}\\ \end{array}$$

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: None

```
\begin{array}{r}{3x=1}\\ {x+2y=3}\end{array}
```

$$\begin{array}{r}{3x=1}\\ {x+2y=3}\end{array}$$

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: None

```
\begin{array}{r}{3x=1}\\ {x+2y=3}\end{array}
```

$$\begin{array}{r}{3x=1}\\ {x+2y=3}\end{array}$$

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

control__lines3

control | native_tex_180dpi | no GT-based routing | all models, same input

ACTUAL INPUT

$$\begin{aligned}x+y&=2\\x-y&=0\\z&=3\end{aligned}$$

```
\begin{aligned}x+y&=2\\x-y&=0\\z&=3\end{aligned}
```

REFERENCE RENDERING

$$\begin{aligned}x+y&=2\\x-y&=0\\z&=3\end{aligned}$$

PP-FormulaNet-S

Strict EM: False | Render: rendered | Matrix cells: None

```
\begin{array}{r}{x+y=2}\\ {x-y=0}\\ {z=3}\end{array}
```

$$\begin{array}{r}{x+y=2}\\ {x-y=0}\\ {z=3}\end{array}$$

PP-FormulaNet_plus-S

Strict EM: False | Render: rendered | Matrix cells: None

```
\begin{aligned}{x+y}&{{}=2}\\ {x-y}&{{}=0}\\ {z}&{{}=3}\\ \end{aligned}
```

$$\begin{aligned}{x+y}&{{}=2}\\ {x-y}&{{}=0}\\ {z}&{{}=3}\\ \end{aligned}$$

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: None

```
\begin{array}{r}{x+y=2}\\ {x-y=0}\\ {z=3}\end{array}
```

$$\begin{array}{r}{x+y=2}\\ {x-y=0}\\ {z=3}\end{array}$$

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: None

```
\begin{array}{r}{x+y=2}\\ {x-y=0}\\ {z=3}\end{array}
```

$$\begin{array}{r}{x+y=2}\\ {x-y=0}\\ {z=3}\end{array}$$

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

control__sup

control | native_tex_180dpi | no GT-based routing | all models, same input

ACTUAL INPUT



```
x^2
```

REFERENCE RENDERING



PP-FormulaNet-S

Strict EM: False | Render: rendered | Matrix cells: None

```
x^{2}
```



PP-FormulaNet_plus-S

Strict EM: False | Render: rendered | Matrix cells: None

```
x^{2}
```



PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: None

```
x^{2}
```



UniMERNet

Strict EM: False | Render: rendered | Matrix cells: None

```
x^{2}
```



Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

control__sub

control | native_tex_180dpi | no GT-based routing | all models, same input

ACTUAL INPUT



REFERENCE RENDERING



```
x_i
```

PP-FormulaNet-S

Strict EM: False | Render: rendered | Matrix cells: None



```
x_{i}
```

PP-FormulaNet_plus-S

Strict EM: False | Render: rendered | Matrix cells: None



```
x_{i}
```

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: None



```
x_{i}
```

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: None



```
x_{i}
```

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

control__sub_sup

control | native_tex_180dpi | no GT-based routing | all models, same input

ACTUAL INPUT

$$x_i^2 + y_j^3$$

```
x_i^2+y_j^3
```

REFERENCE RENDERING

$$x_i^2 + y_j^3$$

PP-FormulaNet-S

Strict EM: False | Render: rendered | Matrix cells: None

```
x_{i}^{2}+y_{j}^{3}
```

$$x_i^2 + y_j^3$$

PP-FormulaNet_plus-S

Strict EM: False | Render: rendered | Matrix cells: None

```
x_{i}^{2}+y_{j}^{3}
```

$$x_i^2 + y_j^3$$

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: None

```
x_{i}^{2}+y_{j}^{3}
```

$$x_i^2 + y_j^3$$

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: None

```
x_{i}^{2}+y_{j}^{3}
```

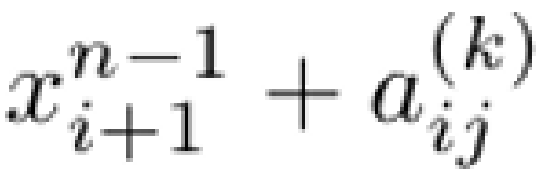
$$x_i^2 + y_j^3$$

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

control__complex_scripts

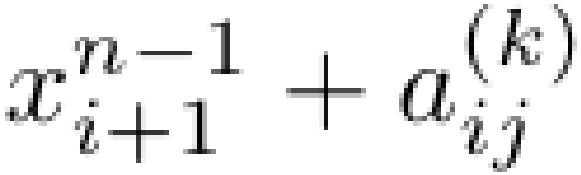
control | native_tex_180dpi | no GT-based routing | all models, same input

ACTUAL INPUT



```
x_{i+1}^{n-1}+a_{ij}^{(k)}
```

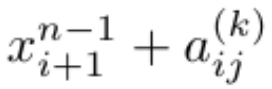
REFERENCE RENDERING



PP-FormulaNet-S

Strict EM: True | Render: rendered | Matrix cells: None

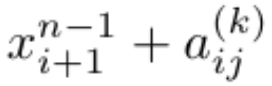
```
x_{i+1}^{n-1}+a_{i j}^{(k)}
```



PP-FormulaNet_plus-S

Strict EM: True | Render: rendered | Matrix cells: None

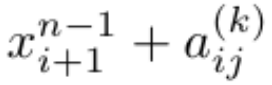
```
x_{i+1}^{n-1}+a_{i j}^{(k)}
```



PP-FormulaNet-L

Strict EM: True | Render: rendered | Matrix cells: None

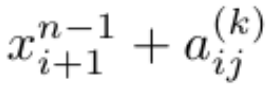
```
x_{i+1}^{n-1}+a_{i j}^{(k)}
```



UniMERNet

Strict EM: True | Render: rendered | Matrix cells: None

```
x_{i+1}^{n-1}+a_{i j}^{(k)}
```



Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

control__set

control | native_tex_180dpi | no GT-based routing | all models, same input

ACTUAL INPUT

$$S = \{1, 2, 3\}$$

```
S=\{1,2,3\}
```

REFERENCE RENDERING

$$S = \{1, 2, 3\}$$

PP-FormulaNet-S

Strict EM: True | Render: rendered | Matrix cells: None

```
S=\{1,2,3\}
```

$$S = \{1, 2, 3\}$$

PP-FormulaNet_plus-S

Strict EM: True | Render: rendered | Matrix cells: None

```
S=\{1,2,3\}
```

$$S = \{1, 2, 3\}$$

PP-FormulaNet-L

Strict EM: True | Render: rendered | Matrix cells: None

```
S=\{1,2,3\}
```

$$S = \{1, 2, 3\}$$

UniMERNet

Strict EM: True | Render: rendered | Matrix cells: None

```
S=\{1,2,3\}
```

$$S = \{1, 2, 3\}$$

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

control__set_text

control | native_tex_180dpi | no GT-based routing | all models, same input

ACTUAL INPUT

$$S_3 = \{A_{3\times3} \mid A \text{ is a permutation matrix}\}$$

```
S_3=\{A_{3\times3}\mid A\text{ is a permutation matrix}\}
```

REFERENCE RENDERING

$$S_3 = \{A_{3\times3} \mid A \text{ is a permutation matrix}\}$$

PP-FormulaNet-S

Strict EM: False | Render: rendered | Matrix cells: None

```
S_{3}=\{A_{3\times3}\mid A{\mathrm{~i s~a~p e r m u t a t i o n~m a t r i x}}\}
```

$$S_3 = \{A_{3\times3} \mid A \mathrm{~i s~a~p e r m u t a t i o n~m a t r i x}\}$$

PP-FormulaNet_plus-S

Strict EM: False | Render: rendered | Matrix cells: None

```
S_{3}=\{A_{3\times3}\mid A{mathrm s a a p e r m u t a t i o n m a t r i x}\}
```

$$S_3 = \{A_{3\times3} \mid A mathrm s a a p e r m u t a t i o n m a t r i x\}$$

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: None

```
S_{3}=\{A_{3\times3}\mid A{\mathrm{~i s~a~p e r m u t a t i o n~m a t r i x}}\}
```

$$S_3 = \{A_{3\times3} \mid A \mathrm{~i s~a~p e r m u t a t i o n~m a t r i x}\}$$

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: None

```
S_{3}=\{A_{3\times3}\mid A{\mathrm{~i s~a~p e r m u t a t i o n~m a t r i x}}\}
```

$$S_3 = \{A_{3\times3} \mid A \mathrm{~i s~a~p e r m u t a t i o n~m a t r i x}\}$$

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.